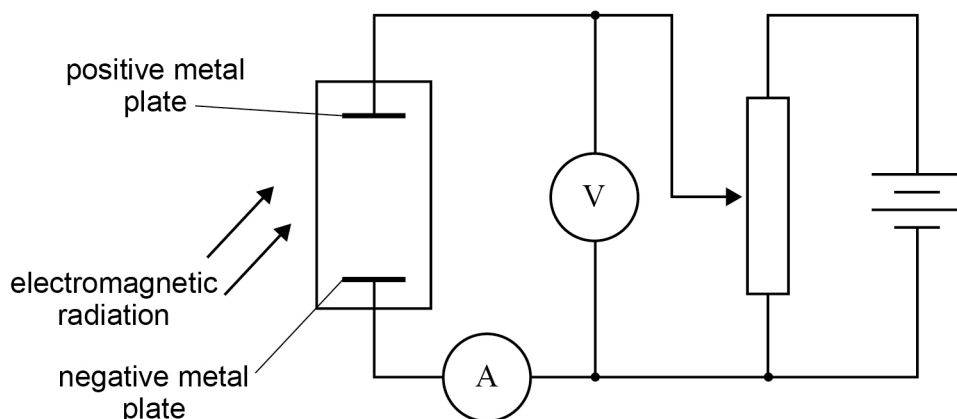


The variation of the maximum kinetic energy of photoelectrons with the wavelength of incident electromagnetic radiation may be investigated using the apparatus in **Figure 11**.

Figure 11



When the positive metal plate is illuminated with monochromatic electromagnetic radiation of a sufficiently high frequency, electrons are emitted and move towards the negative plate, causing a current in the circuit. The potential difference between the positive plate and the negative plate is adjusted until the current is zero. When this happens, the potential difference measured by the voltmeter is called the stopping potential V_s .

The maximum kinetic energy of the photoelectrons $E_{k(\max)}$ is eV_s .

Question 8 continues on the next page

0 8 . 1 In one experiment, the results in **Table 1** were obtained:

Table 1

Frequency of incident radiation / 10^{14} Hz	V_s / V	$E_{k(\max)}$ / 10^{-20} J
6.0	$0.15 \pm$ _____	2.4 ± 0.8
6.8	$0.55 \pm$ _____	8.6 ± 0.8
7.5	$0.80 \pm$ _____	12.8 ± 0.8
8.6	$1.25 \pm$ _____	20.0 ± 0.8

Complete **Table 1** by calculating the values of the absolute uncertainties of V_s .

[1 mark]

0 8 . 2 The values of frequency and $E_{k(\max)}$ have been plotted on **Figure 12**.

Complete the graph by adding error bars for each point and by drawing a best fit straight line.

Assume that there is negligible uncertainty in the values of frequency.

[2 marks]

0 8 . 3 The equation for the graph is:

$$E_{k(\max)} = hf - \phi$$

where h is the Planck constant and ϕ is the work function of the metal from which the positive plate is made.

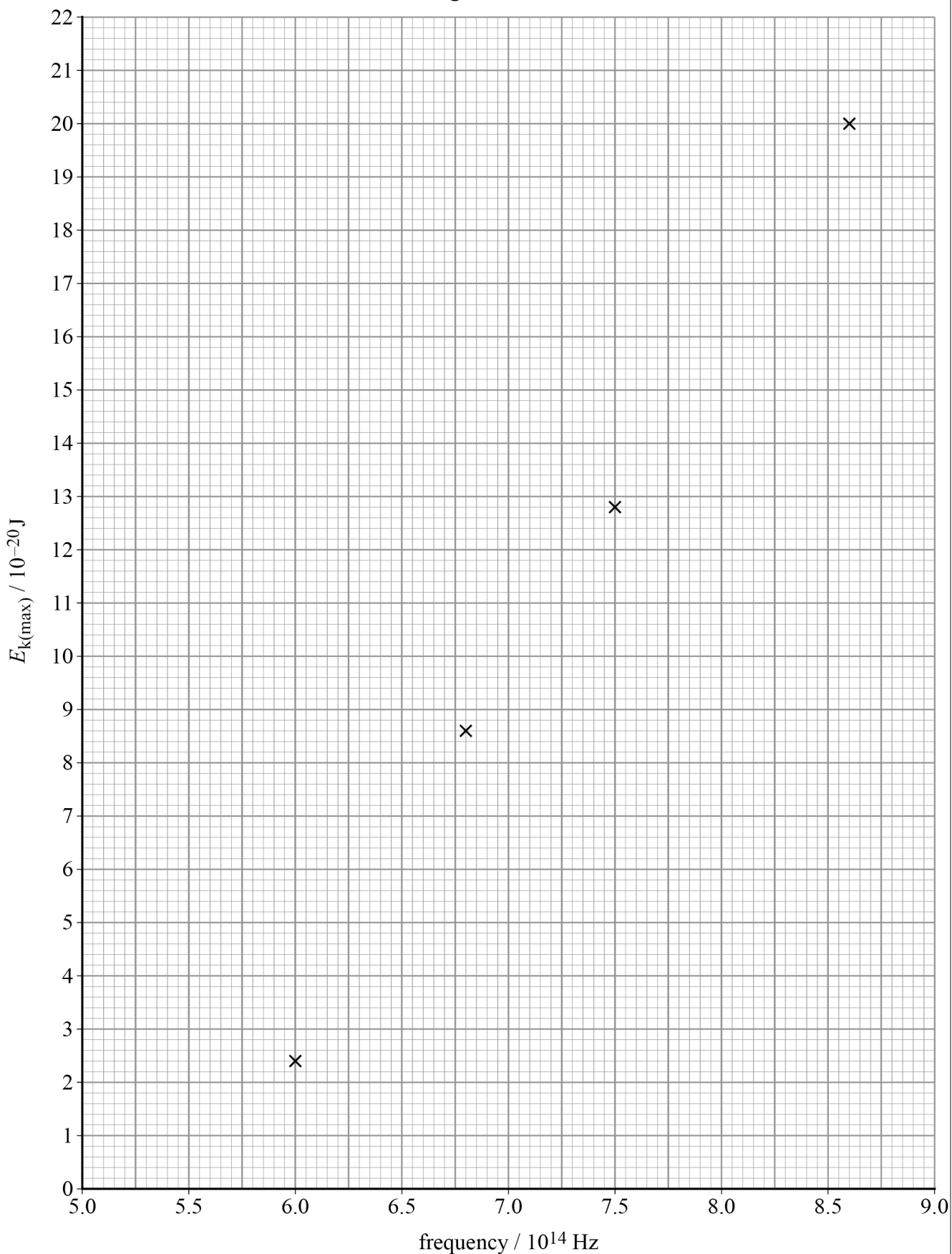
Determine using your graph a value for h .

[2 marks]

$$h = \text{_____ J s}$$



Figure 12



Question 8 continues on the next page

08.4 Photoelectrons are not emitted below a threshold frequency f_0 .

Determine using your graph the value of f_0 .

[1 mark]

$$f_0 = \text{_____ Hz}$$

08.5 Determine using your graph the uncertainty in your value of f_0 .

[2 marks]

$$\text{uncertainty in } f_0 = \pm \text{_____ Hz}$$

08.6 Suggest **one** way in which the results in question **08.1** could be improved.

[1 mark]



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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What is the de Broglie wavelength of an electron travelling at 30% of the speed of light?

Ignore any relativistic effects.

[1 mark]

A $2.4 \times 10^{-14} \text{ m}$

☐

B $8.1 \times 10^{-14} \text{ m}$

☐

C $2.4 \times 10^{-12} \text{ m}$

☐

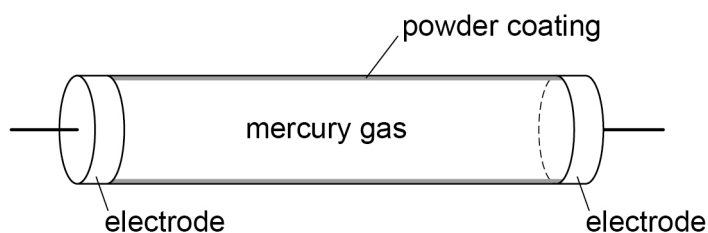
D $8.1 \times 10^{-12} \text{ m}$

☐

Figure 5 shows a fluorescent tube. A fluorescent tube contains low-pressure mercury gas.

The inner surface of the tube is coated with a special powder as shown in **Figure 5**.

Figure 5



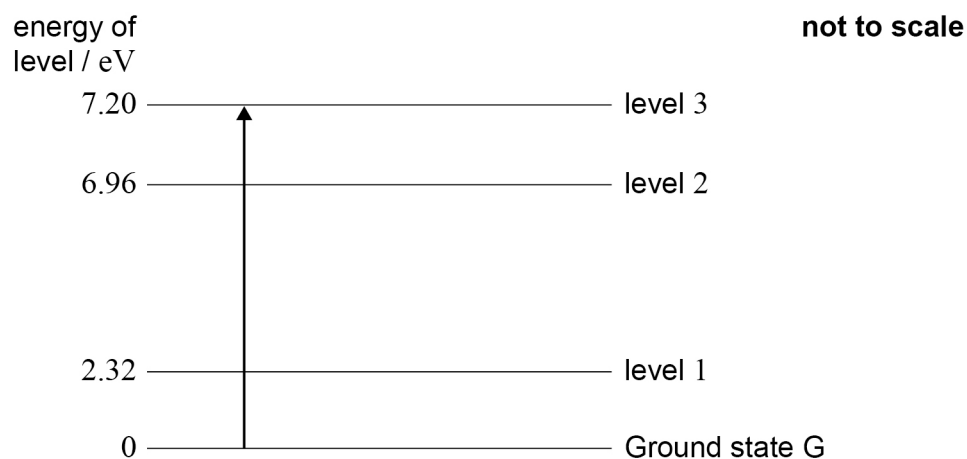
The mercury atoms are excited, and this eventually leads to the excitation of the atoms in the powder.

Explain how the mercury atoms are excited.

[1 mark]

Figure 6 shows the energy levels of one type of atom in the powder.

Figure 6



06.2 Explain how atoms in the powder become excited to the 7.20 eV energy level. **[2 marks]**

06.3 With reference to **Figure 6**, explain how photons of visible light are emitted by the powder. Calculations are not required. **[2 marks]**

06.4 Draw an arrow on **Figure 6** to represent the transition that leads to the emission of the longest wavelength of radiation. **[1 mark]**

06.5 Calculate the **wavelength** of radiation emitted as a result of the transition you identified in question **06.4**. **[3 marks]**

wavelength = _____ m





Figure 1 shows some of the energy levels of a single atom.

Figure 1



0	1	.	1
---	---	---	---

The atom is in its ground state. A photon of energy 8.81 eV is incident on the atom.

Describe a likely outcome of this event.

[2 marks]

0	1	.	2
---	---	---	---

Multiple atoms, with the same energy levels as shown in **Figure 1**, return to the ground state.

State how many different photon wavelengths can be observed.

[1 mark]

number of wavelengths = _____



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1**6**

A point source of gamma radiation emits 1.0×10^5 gamma photons per second.

What is the gamma intensity at a distance of 4.0 m from the source?
Ignore background radiation.

[1 mark]

A 380 photons $\text{m}^{-2} \text{s}^{-1}$

☐

B 500 photons $\text{m}^{-2} \text{s}^{-1}$

☐

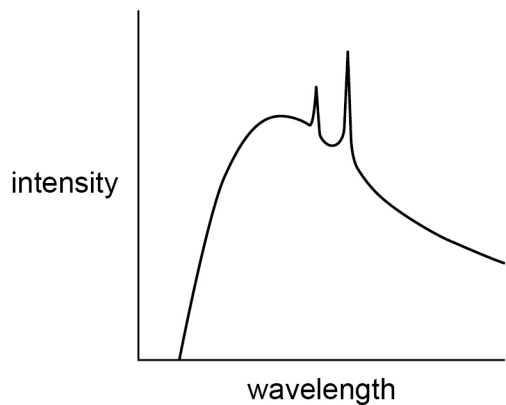
C 6250 photons $\text{m}^{-2} \text{s}^{-1}$

☐

D 25 000 photons $\text{m}^{-2} \text{s}^{-1}$

☐

The graph shows the spectrum of X-rays produced using an anode–cathode potential difference of 30 kV.



Which will **not** increase when the anode–cathode potential difference is increased?

[1 mark]

A the intensity at any point of the continuous spectrum

☐

B the intensity of the characteristic peaks

☐

C the frequency of the characteristic peaks

☐

D the minimum frequency of the continuous spectrum

☐

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2**3**

The photoelectric effect provides evidence that

[1 mark]

A electrons can have wave-like properties.

☐

B electrons always behave as particles.

☐

C electromagnetic waves can have particle-like properties.

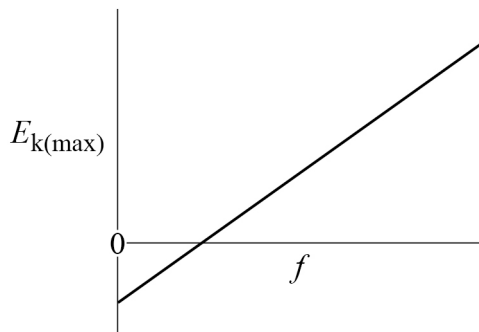
☐

D electromagnetic waves always behave as waves.

☐

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Light is incident on a metal surface. The graph shows the variation of maximum photoelectron kinetic energy $E_{k(\text{max})}$ with the frequency f of incident light in the photoelectric effect.



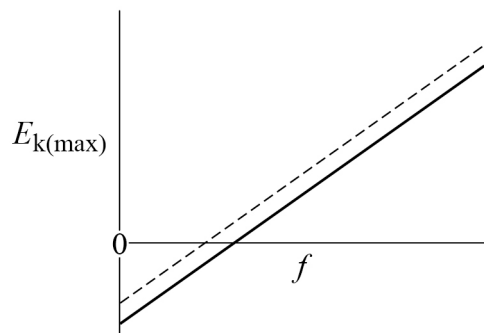
The original line is shown as dotted in all of the graphs on page 31.

Which graph shows the variation of $E_{k(\text{max})}$ with f for a metal with a greater work function?

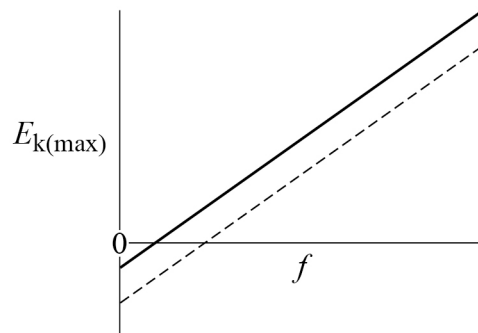
[1 mark]



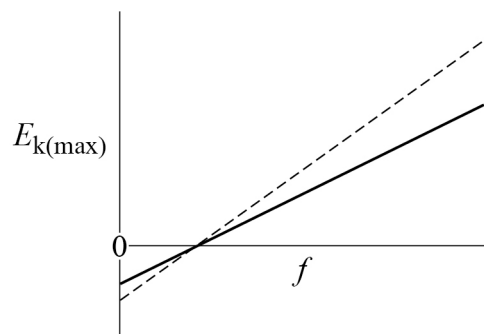
A



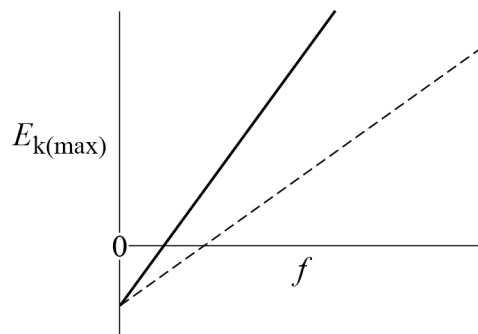
B



C



D



A



B



C



D



[illegible]

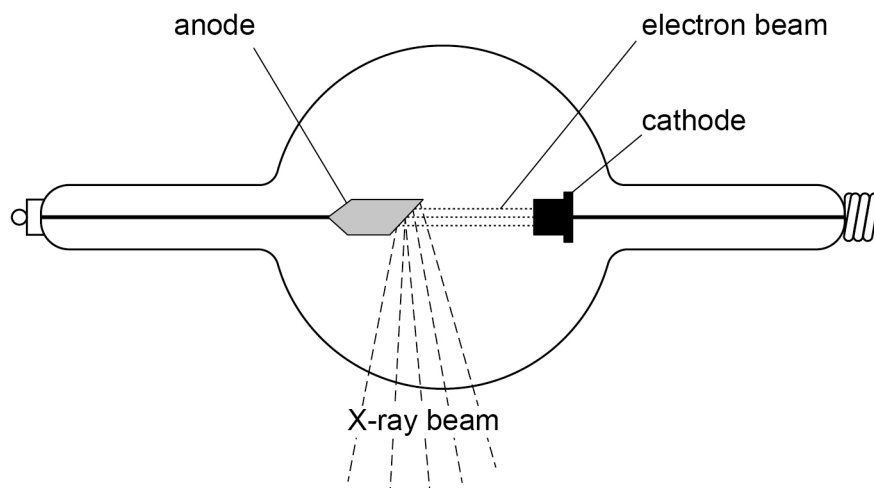
[illegible]

[illegible]

06

Figure 5 shows the main parts of an X-ray tube. Electrons are emitted at the cathode and accelerated onto the anode.

Figure 5



The operating voltage between the anode and the cathode is 80 kV.

The tube current is 100 mA.

The tube is switched on for an operating period of 5.0 ms.

06

.1

Calculate the number of electrons that reach the anode during this operating period.

[2 marks]

number of electrons = _____



0 6 . 2

Energy that is not emitted in the form of photons is transferred to internal energy.
The efficiency of the X-ray tube is 2.0%.

Calculate the increase in internal energy of the anode.

[3 marks]

increase in internal energy = _____ J

0 6 . 3

The tube current is now increased. The operating voltage and the operating period are unchanged.

State and explain any effect that the increased tube current has on the X-rays produced.

[3 marks]



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07

The following equation applies to the photoelectric effect:

$$hf = \phi + E_{k(\max)}$$

07.1

State what quantities are represented by the terms hf and $E_{k(\max)}$.

[2 marks]

hf _____

$E_{k(\max)}$ _____

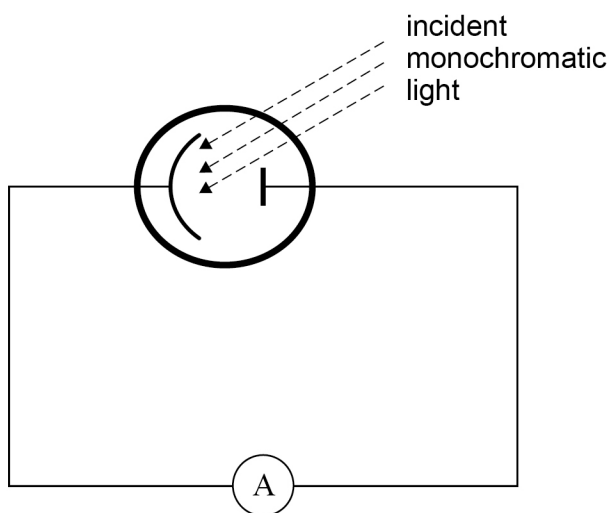
07.2

State and explain what is represented by the term ϕ .

[2 marks]

Figure 6 shows a circuit used to investigate the photoelectric effect. The ammeter can measure very small currents.

Figure 6



Monochromatic red light is incident on the metal surface of the cathode and the current in the ammeter is zero.

The red light is replaced with monochromatic blue light and there is a current in the ammeter.

0 7 . 3

Explain why there is a current with blue light but not with red light.

[2 marks]

0 7 . 4

The intensity of the monochromatic blue light is increased and the experiment is repeated.

State and explain any change in the current.

[3 marks]



Do not write

2 5

An alpha particle moving at a speed v has de Broglie wavelength λ .

What is the de Broglie wavelength of a proton moving at a speed $2v$?

[1 mark]

A $\frac{\lambda}{4}$



B $\frac{\lambda}{2}$



C 2λ



D 4λ



[illegible]

[illegible]

[illegible]

0	1
---	---

A gamma photon has a frequency of 3.3×10^{29} Hz.

Calculate, in TeV, the energy of this photon.

[2 marks]

energy of photon = _____ TeV

2

Photoelectrons are emitted when monochromatic light is incident on a metal surface. The emitted photoelectrons have a range of kinetic energies up to a maximum value.

Explain why this energy range occurs.

[3 marks]

3

Turn over for the next question

Turn over ►



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Explain why the spectrum from an X-ray tube has characteristic lines.

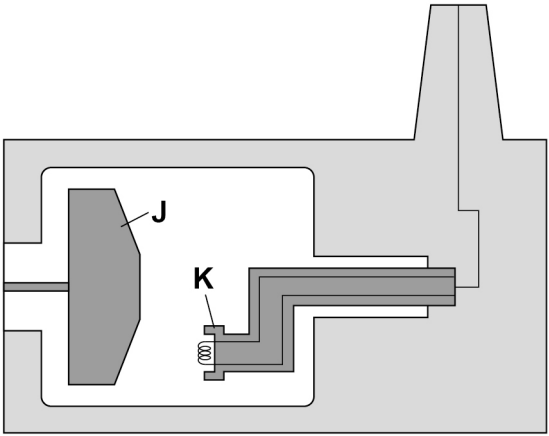
[3 marks]

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The diagram shows an X-ray tube.



Which row identifies the anode, the cathode and the target?

[1 mark]

	Anode	Cathode	Target	
A	J	K	J	☐
B	K	J	J	☐
C	J	K	K	☐
D	K	J	K	☐

Turn over for the next question



Do not write

1

9

A nucleus of hydrogen ${}^2_1\text{H}$ is moving at a constant speed.

The de Broglie wavelength of the nucleus is $2.2 \times 10^{-10} \text{ m}$.

What is an estimate of the kinetic energy of the nucleus?

[1 mark]

A $7.5 \times 10^{-25} \text{ J}$

☐

B $1.5 \times 10^{-24} \text{ J}$

☐

C $1.4 \times 10^{-21} \text{ J}$

☐

D $2.7 \times 10^{-21} \text{ J}$

☐

Monochromatic light is incident on a clean metal surface and photoelectrons are emitted from the surface.

A photon of this light has energy of $4.5 \times 10^{-19} \text{ J}$.

The stopping potential of the electrons is 1.1 V .

Calculate, in eV, the work function of the metal.

[2 marks]

work function = _____ eV



Don't write

0	6
---	---

A sodium lamp contains low-pressure sodium gas.
An electric current in the sodium gas causes the sodium atoms to emit light.

0	6	.	1
---	---	---	---

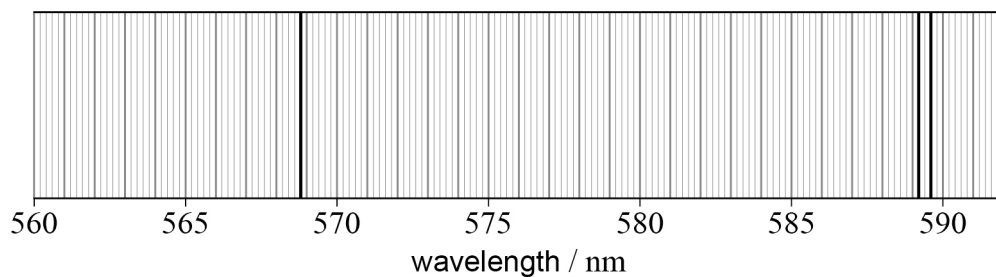
Explain how the sodium atoms become excited.

[2 marks]



Figure 6 shows part of the line spectrum for the light emitted from the sodium lamp.

Figure 6



0 6 . 2

Explain why the spectrum produced by the sodium lamp has light of only certain wavelengths.

[3 marks]

0 6 . 3

Calculate, in J, the energy of the **least** energetic photon that corresponds to a spectral line in **Figure 6**.

[3 marks]

photon energy = _____ J



Do not write

A particle travels at a speed of $3.6 \times 10^6 \text{ m s}^{-1}$.

The de Broglie wavelength of the particle is 0.98 pm .

Calculate the mass of the particle.

[2 marks]

mass of particle = _____ kg

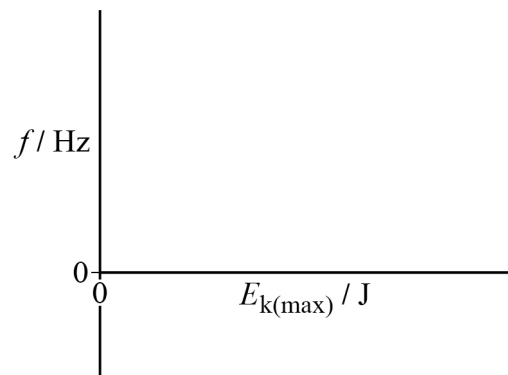
2



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1 3

Light of frequency f is incident on a metal surface.
 Photoelectrons with maximum kinetic energy $E_{k(\max)}$ are emitted.
 A graph showing the variation of f with $E_{k(\max)}$ is plotted using the axes below.



Which row shows how the Planck constant and the work function of the metal surface are determined?
 [1 mark]

	Planck constant	Work function	
A	gradient	$\frac{f \text{ intercept}}{\text{gradient}}$	☐
B	gradient	$f \text{ intercept} \times \text{gradient}$	☐
C	$\frac{1}{\text{gradient}}$	$\frac{f \text{ intercept}}{\text{gradient}}$	☐
D	$\frac{1}{\text{gradient}}$	$f \text{ intercept} \times \text{gradient}$	☐

Turn over for the next question



Don't write

An atom is in its ground state.

The ionisation energy of the atom is the minimum energy required to:

[1 mark]

A remove an electron from the atom.

☐

B remove all electrons from the atom.

☐

C add an electron to the atom.

☐

D move an electron within the atom so that the atom's energy level increases.

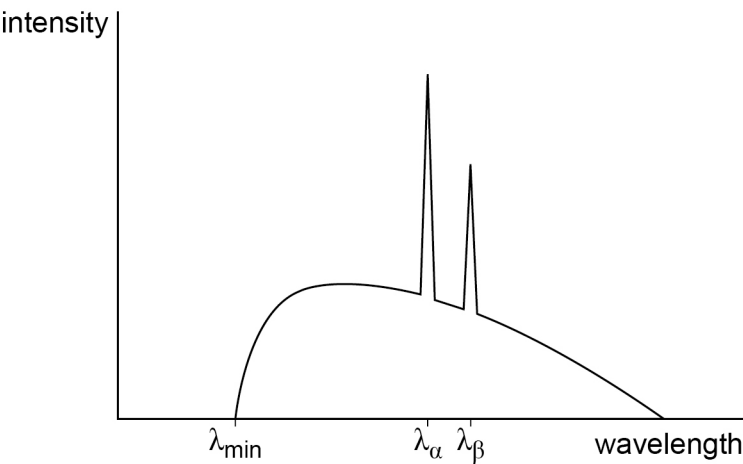
☐

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The graph shows the emission spectrum for an X-ray tube operating at 80 kV.

Characteristic lines occur at wavelengths λ_α and λ_β .

The shortest wavelength in the emission spectrum is $\lambda_{\min}$.



The same X-ray tube is now operated at 40 kV.

What happens to λ_α , λ_β and $\lambda_{\min}$?

[1 mark]

	λ_α	λ_β	$\lambda_{\min}$
A	doubled	doubled	doubled
B	halved	halved	halved
C	unchanged	unchanged	halved
D	unchanged	unchanged	doubled

☐☐☐☐

Turn over for the next question



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01

When light of frequency 9.1×10^{14} Hz is incident on a metal surface, electrons are emitted. The stopping potential for these electrons is 1.65 V.

01.1

State what is meant by the threshold frequency for a metal.

[1 mark]

01.2

Calculate the threshold frequency for this metal.

[3 marks]

threshold frequency = _____ Hz



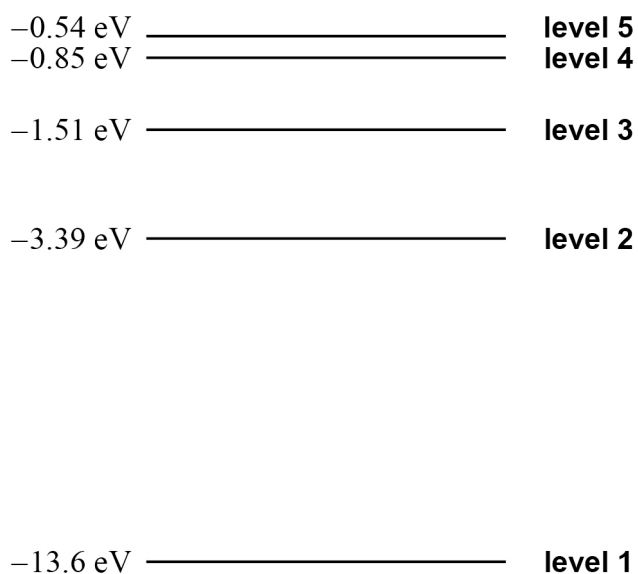
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09.1

Explain how line spectra provide evidence that gas atoms have discrete energy levels.
[3 marks]

Figure 12 shows the lowest energy levels of atomic hydrogen.

Figure 12



0 9 . 2 Determine the transition that emits light with a wavelength of 656 nm.

[4 marks]

transition from level _____ to level _____

0 9 . 3 A beam of photons, each of energy 12.75 eV, is incident on hydrogen atoms that are in level 1.

Determine the number of different photon energies that are emitted by the hydrogen atoms.

[1 mark]

number = _____

8

END OF SECTION A

Turn over ►



Section B

Answer **all** questions in this section.

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The purpose of the powder coating in a fluorescent tube is to absorb one type of particle and emit another type of particle.

Which row is correct?

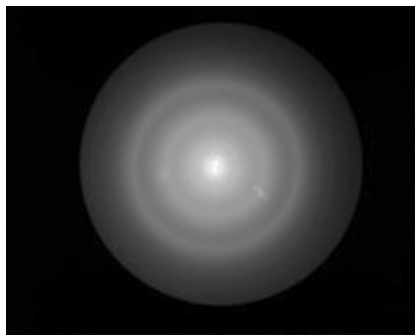
[1 mark]

	Particle absorbed by the powder	Particle emitted by the powder	
A	ultraviolet photons	electrons	☐
B	ultraviolet photons	visible photons	☐
C	electrons	visible photons	☐
D	electrons	ultraviolet photons	☐



Do not write

A beam of electrons is diffracted, creating a diffraction pattern consisting of bright circles.



The momentum of each electron is then decreased.

The number of electrons emitted per second in the beam stays the same.

Which row shows what happens to the de Broglie wavelength of the electrons and the diameter of the bright circles?

[1 mark]

	de Broglie wavelength	Diameter	
A	increases	decreases	☐
B	increases	increases	☐
C	decreases	decreases	☐
D	decreases	increases	☐

[illegible]



An electron is accelerated from rest through a potential difference V .

The electron now has a de Broglie wavelength of 1.5 nm.

Calculate V .

[3 marks]

$V =$ _____ V

3

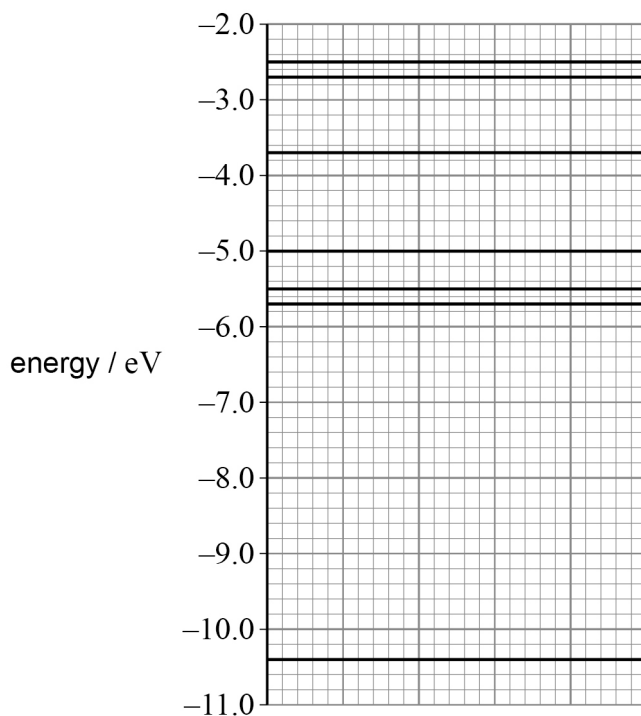


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07

Figure 6 shows some of the energy levels in an atom of mercury.

Figure 6



07.1

Explain why the energy levels are negative.

[2 marks]

07.2

Draw an arrow on **Figure 6** to represent the transition that leads to the emission of the **shortest** wavelength of light for these energy levels.

[1 mark]



0 7 . 3

Calculate the wavelength that is consistent with the transition you chose in Question **07.2**.

[3 marks]

wavelength = _____ m

A fluorescent tube contains a low-pressure mercury gas.

0 7 . 4

Describe how the mercury atoms become excited in the fluorescent tube.

[2 marks]

0 7 . 5

Explain how the excitation of mercury atoms leads to the emission of white light from the fluorescent tube.

[3 marks]



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2

2

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Electrons are accelerated through a potential difference of 6 kV and then directed at a metal target.

Which is **not** a possible consequence of the electrons hitting the target?

[1 mark]

A X-rays are produced.

☐

B Photoelectrons are emitted.

☐

C The temperature of the target increases.

☐

D Photons are emitted.

☐

2**3**

Which can ionise an atom?

[1 mark]**A** fast-moving electrons☐**B** infrared radiation☐**C** microwaves☐**D** neutrinos☐**Turn over for the next question****Turn over ►**

Do not write

Monochromatic light from a source **L1** is shone on a metal surface, causing electrons to be emitted.

The light has frequency f and the power of **L1** is P .

The stopping potential of the electrons and the rate of emission of the electrons are measured.

L1 is replaced with a different monochromatic light source **L2**. The stopping potential of the electrons is unchanged and the rate of emission of the electrons doubles.

Which row is correct?

[1 mark]

	Frequency of light from L2	Power of L2	
A	f	$2P$	☐
B	$2f$	P	☐
C	$2f$	$2P$	☐
D	$\frac{f}{2}$	P	☐

2**5**

Which statement is correct?

[1 mark]

A The photoelectric effect demonstrates the wave-like behaviour of electrons.

☐

B The photoelectric effect demonstrates the particle-like behaviour of light.

☐

C The diffraction of electrons demonstrates the particle-like behaviour of electrons.

☐

D The diffraction of electrons demonstrates the wave-like behaviour of light.

☐

14

[illegible]

[illegible]

[illegible]

[illegible]